

The rkdiagram Package

Reed-Kellogg Sentence Diagrams for $\text{\LaTeX}$

Version 1.3 • Requires $\text{\LaTeX}$

Introduction

The `rkdiagram` package typesets traditional Reed-Kellogg sentence diagrams in $\text{\LaTeX}$. Tag each word with a command describing its grammatical role; the package computes the layout and draws everything with `TikZ`.

Features: baselines, dividers, modifier diagonals, prepositional phrases, appositives, compound elements (including compound objects of prepositions), subordinate clauses, compound sentences, gerunds, and participles.

Requirements: $\text{\LaTeX}$ plus `tikz`, `xparse`, `etoolbox`, and `pgfopts` — all in a standard TeX Live or MiKTeX installation.

Loading the Package

```
\usepackage[
  spacing      = 1.3cm,
  modDepth     = 1.0cm,
  modgap       = 0.25cm,
  cmpSep       = 0pt,
  subClauseDepth = 2.5cm,
  linewidth    = 0.5pt,
  font         = \rmfamily,
  color        = black,
]{rkdiagram}
```

All options are optional; the values above are the defaults. Override any option mid-document with `\rkset`:

```
\rkset{spacing=1.5cm, color=blue}
```

Package options

Option	Default	Effect
spacing	1.2cm	Horizontal gap (padding) between baseline elements
modDepth	1.0cm	Fixed horizontal step between plain adj/adv modifiers; minimum slant length for
modgap	0.25cm	Extra vertical headspace between nested modifier levels
cmpSep	0pt	Minimum half-gap between compound platforms (0pt = auto)
subClauseDepth	2.5cm	Vertical drop to a subordinate clause baseline
gerundPost	1.2cm	Minimum post height for gerund T-platforms
linewidth	0.4pt	Thickness of all diagram lines
font	(doc font)	Font applied to all diagram words
color	black	Line and text colour
bridgeY	0pt	Pin compound-sentence bridge y (0pt = auto)

The rkdiagram Environment

Every diagram lives inside `rkdiagram`, which opens a *TikZ* picture:

```
\begin{rkdiagram}[spacing=1.5cm] % optional per-diagram options
  \rkSubject{word}
  \rkVerb{word}
\end{rkdiagram}
```

Words are automatically centred on their baseline lines. The diagram sits on the text baseline and can appear inline or in its own paragraph.

Baseline Elements

Baseline elements sit on the main horizontal line, placed left to right in the order written.

Subject and Verb

A full vertical bar separates subject from verb. The bar's ascender (above the baseline) is full height; its descender (below the baseline) is half that height.

“Dogs bark.”

```
\begin{rkdiagram}
  \rkSubject{Dogs}
  \rkVerb{bark}
\end{rkdiagram}
```



Direct Object

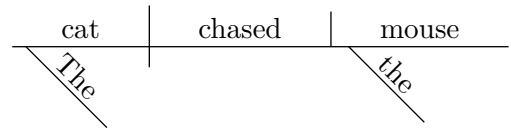
`\rkDObj` places a short bar (above baseline only) after the verb.

“The cat chased the mouse.”

```

\begin{rkdiagram}
  \rkSubject[\rkAdj{The}]{cat}
  \rkVerb{chased}
  \rkDObj[\rkAdj{the}]{mouse}
\end{rkdiagram}

```



Indirect Object

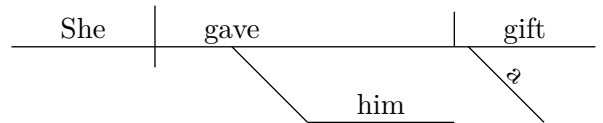
`\rkIObj` draws a diagonal platform below the verb midpoint.

"She gave him a gift."

```

\begin{rkdiagram}
  \rkSubject{She}
  \rkVerb{gave}
  \rkIObj{him}
  \rkDObj[\rkAdj{a}]{gift}
\end{rkdiagram}

```



Predicate Nominative and Predicate Adjective

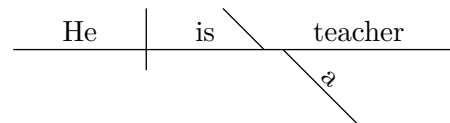
After a linking verb, a back-diagonal divider (`\`) introduces a predicate nominative (`\rkPredNom`) or predicate adjective (`\rkPredAdj`). The diagonal slants at 45°, matching the modifier diagonals, and its bottom touches the baseline without descending below it.

"He is a teacher."

```

\begin{rkdiagram}
  \rkSubject{He}
  \rkVerb{is}
  \rkPredNom[\rkAdj{a}]{teacher}
\end{rkdiagram}

```



Modifiers

Modifiers hang below the word they modify on a 45° diagonal. Use `\rkAdj` for adjectives and articles, `\rkAdv` for adverbs. Pass them as the *optional first argument* of the word they modify, comma-separated:

```

\rkSubject[\rkAdj{The}, \rkAdj{old}]{dog}

```

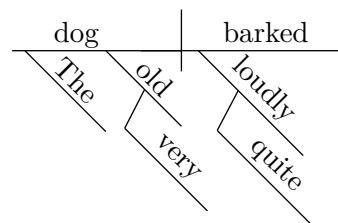
A modifier takes its own optional argument for sub-modifiers, creating chains of diagonals:

"The very old dog barked quite loudly."

```

\begin{rkdiagram}
  \rkSubject[
    \rkAdj{The},
    \rkAdj[\rkAdv{very}]{old}
  ]{dog}
  \rkVerb[\rkAdv[\rkAdv{quite}]{loudly}]{barked}
\end{rkdiagram}

```



The `modgap` option adds extra vertical clearance between nested levels (default 0.25 cm), preventing crowding when modifier chains are deep.

Modifier spacing: plain adjectives and adverbs (no sub-modifiers) advance the sibling cursor by exactly `modDepth`, giving even horizontal spacing regardless of label width. Their slant lines are still sized to fit the word — long labels simply extend past the slant ends, matching hand-drawn convention. Modifiers that carry sub-modifiers (e.g. `\rkAdj[\rkAdv{very}]{old}`) use the word-fit formula so the slant is long enough to keep the sub-modifier connector clear of the next sibling. Use `modDepth` to control the global step between plain modifiers.

Prepositional Phrases

`\rkPrep` draws a diagonal with the preposition label, then a horizontal line for the object.

```
\rkPrep[mods-on-object]{preposition}{object}
```

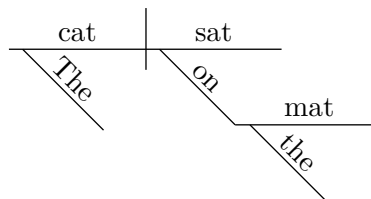
Place it in the modifier list of the word it modifies:

“The cat sat on the mat.”

```

\begin{rkdiagram}
  \rkSubject[\rkAdj{The}]{cat}
  \rkVerb[\rkPrep[\rkAdj{the}]{on}{mat}]{sat}
\end{rkdiagram}

```



Prepositional phrases nest freely — a prep modifying the object of another prep:

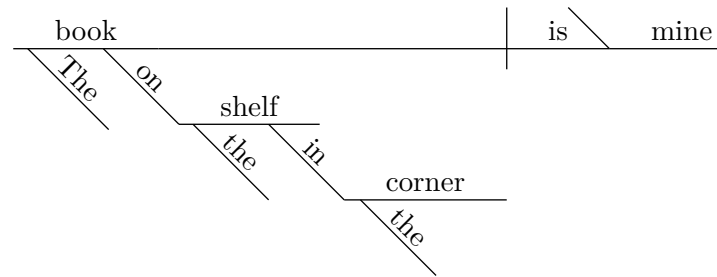
“The book on the shelf in the corner is mine.”

```

\begin{rkdiagram}
  \rkSubject[
    \rkAdj{The},
    \rkPrep[
      \rkAdj{the},
      \rkPrep[\rkAdj{the}]{in}{corner}
    ]{on}{shelf}
  ]{book}

  \rkVerb{is}
  \rkPredNom{mine}
\end{rkdiagram}

```



Compound Structures

Basic usage

`\rkCompound` joins two words or phrases with a conjunction. The full signature is:

```
\rkCompound[conj][mods_a]{word_a}[mods_b]{word_b}
```

- `[conj]` — conjunction label (default: *and*)
- `[mods_a]` / `[mods_b]` — optional per-word modifier lists (same syntax as baseline commands)
- `{word_a}` / `{word_b}` — the two words or phrases

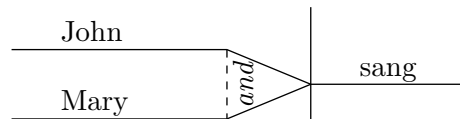
Each word sits on its own horizontal platform. The platforms are connected by convergence slants meeting at a point, a dashed vertical bar at the junction, and the conjunction written between the bar and the convergence point. Vertical separation scales with the conjunction width automatically.

“John and Mary sang.”

```

\begin{rkdiagram}
  \rkSubject{\rkCompound[and]{John}{Mary}}
  \rkVerb{sang}
\end{rkdiagram}

```



Per-word modifiers

Modifiers on each compound word go in the optional modifier-list arguments:

“He painted the wall but not the ceiling.”

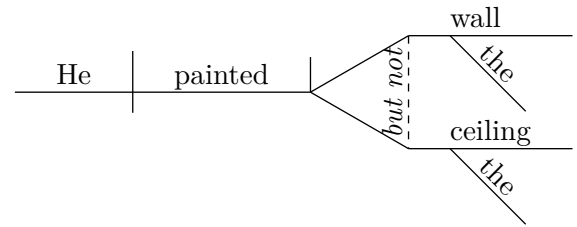
```

\begin{rkdiagram}
  \rkSubject{He}

  \rkVerb{painted}
  \rkDObj{
    \rkCompound[but not]
      [\rkAdj{the}]{wall}

      [\rkAdj{the}]{ceiling}
  }
\end{rkdiagram}

```



Modifiers placed in the list (e.g. `\rkAdj`, `\rkAdv`, `\rkDObj`) draw as proper diagonals on each word's platform. Slot commands like `\rkDObj` inside a per-word modifier list attach a direct object to that compound word's platform line.

Compound object of a preposition

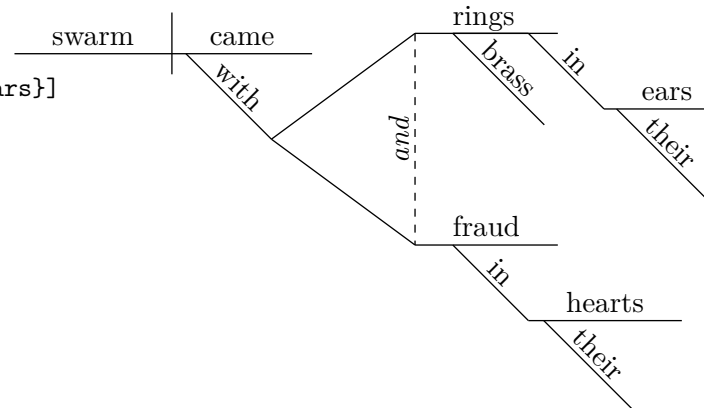
Pass `\rkCompound` as the object argument of `\rkPrep` to fork the preposition into two objects:

"...with brass rings in their ears and fraud in their hearts..."

```

\rkPrep{with}{
  \rkCompound[and]
    [\rkAdj{brass}, \rkPrep[\rkAdj{their}]{in}{ears}]
    {rings}
    [\rkPrep[\rkAdj{their}]{in}{hearts}]
    {fraud}
}

```



Use `cmpSep` when modifiers on the compound words are deep and need extra vertical room between platforms.

Compound Sentences

Use `\rkCompoundSent` *standalone* (not inside `rkdiagram`) to diagram two independent clauses in a single picture, connected by a dashed-line bridge with the coordinating conjunction. Each clause body must be wrapped in its own `rkdiagram` environment:

```

\rkCompoundSent[conj][sep][opts1]
  {\begin{rkdiagram}body1\end{rkdiagram}}

```

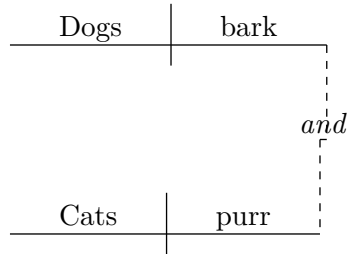
```
[opts2]
{\begin{rkdiagram}body2\end{rkdiagram}}
```

- [conj] — conjunction (default: *and*)
- [sep] — vertical distance between the two baselines (default: 3 cm)
- [opts1] / [opts2] — optional per-clause `\rkset`-style options
- {body1} / {body2} — full `rkdiagram` environments for each clause

The connector drops a dashed vertical from each clause’s verb to a midpoint bridge, draws a short horizontal line there, and labels it with the conjunction.

“Dogs bark and cats purr.”

```
\rkCompoundSent[and][2.5cm]
{\begin{rkdiagram}
  \rkSubject{Dogs}
  \rkVerb{bark}
\end{rkdiagram}}
{\begin{rkdiagram}
  \rkSubject{Cats}
  \rkVerb{purr}
\end{rkdiagram}}
```

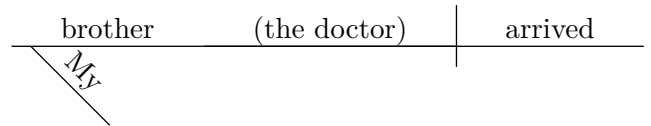


Appositives

`\rkAppos` places an appositive in parentheses on the baseline, directly to the right of the word it renames. Modifier commands inside `\rkAppos` render inline.

“My brother, the doctor, arrived.”

```
\begin{rkdiagram}
  \rkSubject[
    \rkAdj{My},
    \rkAppos{\rkAdj{the} doctor}
  ]{brother}
  \rkVerb{arrived}
\end{rkdiagram}
```



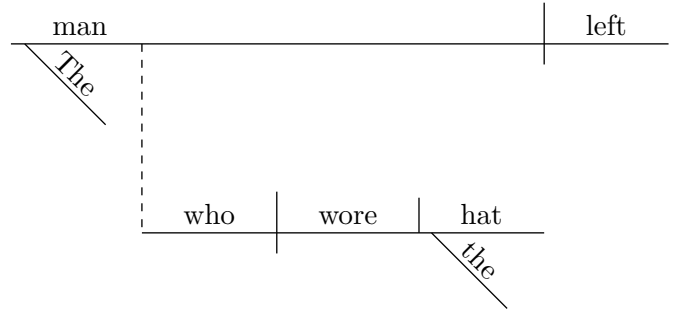
Subordinate Clauses

`\rkSubClause` draws a complete sub-baseline below the main one, connected by a dashed vertical. It appears in the modifier list of the word it modifies. The optional argument is the relative or subordinating word placed first on the sub-baseline; it doubles as the grammatical subject, so do *not* add a separate `\rkSubject` call for it.

```
\rkSubClause[rel-word]{body}
```

“The man who wore the hat left.”

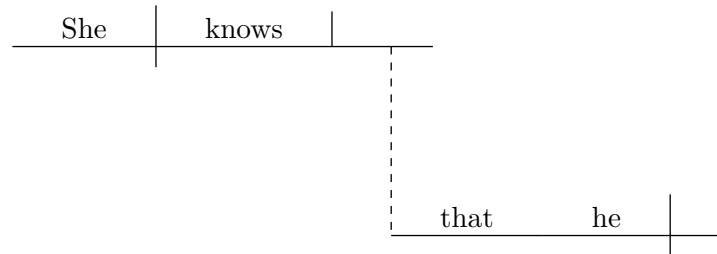
```
\begin{rkdiagram}
  \rkSubject[
    \rkAdj{The},
    \rkSubClause[who]{
      \rkVerb{wore}
      \rkDObj[\rkAdj{the}]{hat}
    }
  ]{man}
  \rkVerb{left}
\end{rkdiagram}
```



A noun clause as direct object — the sub-clause body starts directly with `\rkVerb`:

“She knows that he lied.”

```
\begin{rkdiagram}
  \rkSubject{She}
  \rkVerb{knows}
  \rkDObj{
    \rkSubClause[that]{
      \rkSubject{he}
      \rkVerb{lied}
    }
  }
\end{rkdiagram}
```



Gerunds

A *gerund* is a verb form used as a noun. It sits on a stepped T-platform: a full-width horizontal foot at the noun-slot baseline, two short angled legs rising to an apex, a vertical post, and a short word-platform at the top. Use `\rkGerund` as the *word argument* of any baseline command (`\rkSubject`, `\rkDObj`, `\rkPredNom`, etc.) or as the object of `\rkPrep`.

```
\rkGerund[mods]{word}
```

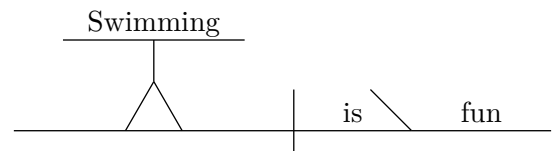
The optional `[mods]` argument holds modifiers of the gerund itself (adjectives, prepositional phrases, etc.). The package pre-scans the list to compute a post height tall enough to keep all modifier slants clear of the foot. Use the `gerundPost` option (default 1.2cm) to set a minimum post height globally.

“Swimming is fun.”


```

\begin{rkdiagram}
\rkSubject{
\rkGerund[\rkPredAdj{fun}]
}
\end{rkdiagram}

```



```
\end{rkdiagram}
```

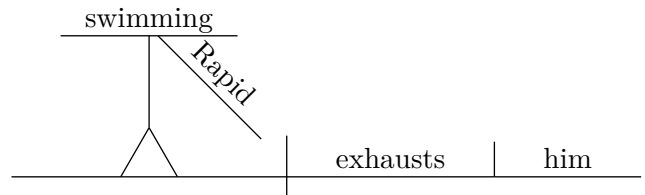
Gerund with its own modifiers — the post height adjusts automatically:

“Rapid swimming exhausts him.”

```

\begin{rkdiagram}
\rkSubject{
\rkGerund[\rkAdj{Rapid}]{swimming}
}
\rkVerb{exhausts}
\rkDObj{him}
\end{rkdiagram}

```



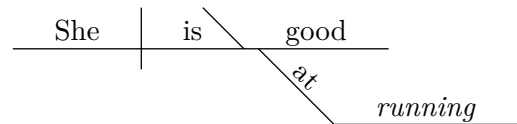
Gerund as the object of a preposition:

“She is good at running quickly.”

```

\begin{rkdiagram}
\rkSubject{She}
\rkVerb{is}
\rkPredAdj[
\rkPrep{at}{\rkGerund[\rkAdv{quickly}]{running}}
]{good}
\end{rkdiagram}

```



Participles

A *participle* functions as an adjective modifying a noun. It is drawn with a hockey-stick shape: a 45° diagonal (word label above it), a short shallower transition, and a horizontal stub. Use `\rkParticiple` inside a modifier list — the same position as `\rkAdj`.

```
\rkParticiple[tail]{word}
```

Without a tail argument the shape is simply the angled line with word label — enough for a bare participial adjective. With a tail, the horizontal stub serves as a mini-baseline for a participial direct object and/or modifiers of the participle:

```

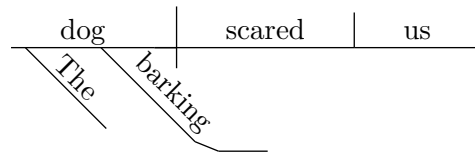
% Participial object on the tail:
\rkPartObj[mods]{word}

```

```
% Adverb or prep phrase modifying the participle, also on the tail:
\rkAdv{quickly}
\rkPrep[obj-mods]{prep}{obj}
```

“The barking dog scared us.”

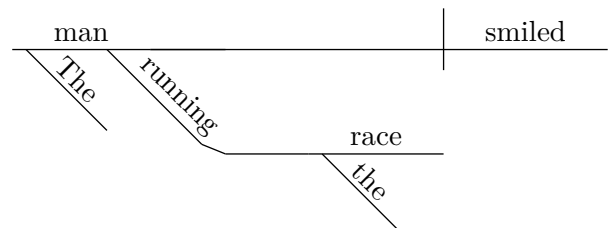
```
\begin{rkdiagram}
\rkSubject[
  \rkAdj{The},
  \rkParticiple{barking}
]{dog}
\rkVerb{scared}
\rkDObj{us}
\end{rkdiagram}
```



Participle with a participial object (`\rkPartObj`) on the tail:

“The man running the race smiled.”

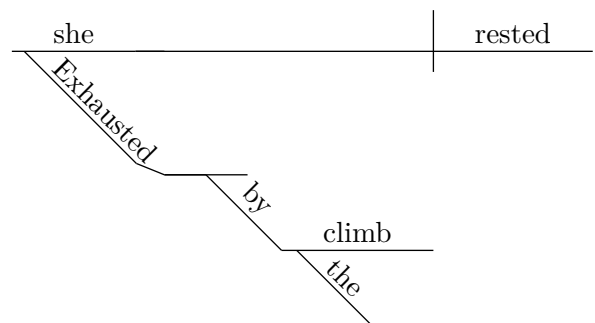
```
\begin{rkdiagram}
\rkSubject[
  \rkAdj{The},
  \rkParticiple[
    \rkPartObj[\rkAdj{the}]{race}
  ]{running}
]{man}
\rkVerb{smiled}
\end{rkdiagram}
```



Participle with a prepositional phrase on the tail (the traditional passive-participial form):

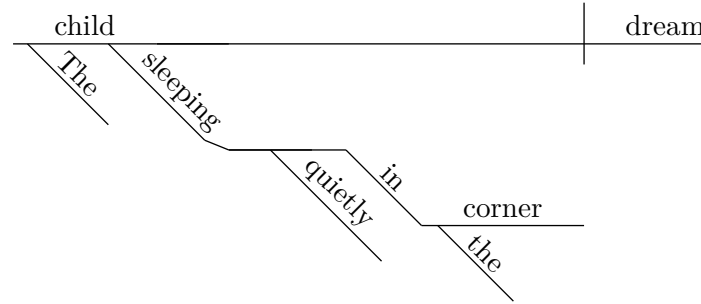
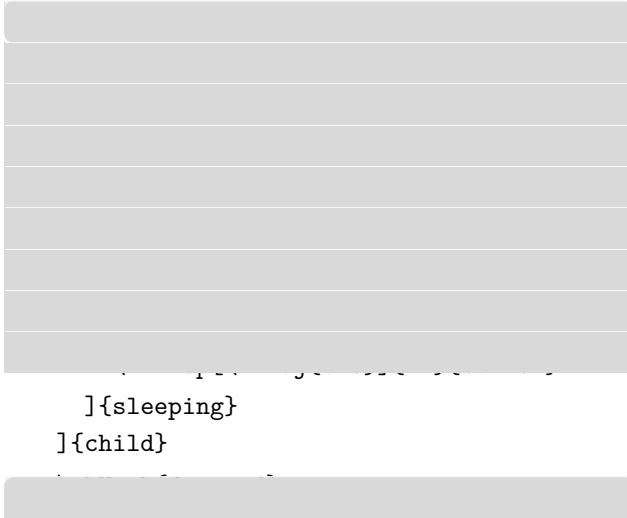
“Exhausted by the climb, she rested.”

```
\begin{rkdiagram}
\rkSubject[
  \rkParticiple[
    \rkPrep[\rkAdj{the}]{by}{climb}
  ]{Exhausted}
]{she}
\rkVerb{rested}
\end{rkdiagram}
```



Multiple tail items (adverb + prepositional phrase) both branch from the horizontal stub:

“The child sleeping quietly in the corner dreamed.”



Manual Spacing Controls

Three one-shot commands let you override automatic spacing for a single element; they reset to zero automatically after use.

```

\rkSlotWidth{len}    % min line width for the NEXT baseline slot
\rkSkip{len}         % gap in baseline BEFORE the next slot
\rkSlantLen{len}     % min diagonal length for the NEXT modifier/prep

```

`\rkSlotWidth` — minimum word line length

Forces the next slot’s baseline line to be at least `len` wide, regardless of word width. Useful for aligning parallel diagrams or leaving annotation space.

“Dogs bark. (4 cm subject line)”

```

\begin{rkdiagram}
\rkSlotWidth{4cm}
\rkSubject{Dogs}
\rkVerb{bark}
\end{rkdiagram}

```

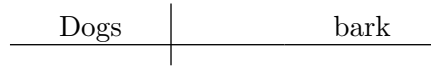


`\rkSkip` — gap before a slot

Inserts a gap of `len` in the baseline *before* the next slot. The gap is drawn as a continuation of the current baseline line, effectively pushing everything right from that point.

“Dogs bark. (1.5 cm gap before verb)”

```
\begin{rkdiagram}
  \rkSubject{Dogs}
  \rkSkip{1.5cm}
  \rkVerb{bark}
```



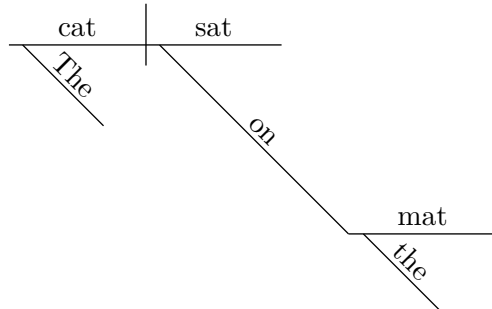
```
\end{rkdiagram}
```

`\rkSlantLen` — minimum diagonal length

Forces the next modifier or prepositional slant to be at least `len` long. Place it on the line immediately before the `\rkAdj`, `\rkAdv`, or `\rkPrep` call.

“The cat sat on the mat. (longer prep slant)”

```
\begin{rkdiagram}
  \rkSubject[\rkAdj{The}]{cat}
  \rkVerb[
    \rkSlantLen{2.5cm}
    \rkPrep[\rkAdj{the}]{on}{mat}
  ]{sat}
\end{rkdiagram}
```



All three commands work inside sub-clauses and on compound-word platforms as well as the main baseline.

Structural Depth Controls

The global options `modDepth`, `subClauseDepth`, and `cmpSep` control the vertical geometry of the diagram. Any of them can be changed per-diagram by passing options to the `rkdiagram` environment, or mid-document with `\rkset`:

```
\begin{rkdiagram}[modDepth=1.5cm, subClauseDepth=3cm]
  ...
\end{rkdiagram}
```

`modDepth` — modifier spacing and slant depth

Controls two related things simultaneously. For plain adjectives and adverbs (no sub-modifiers), it is the *fixed horizontal step* between siblings — all such modifiers are evenly spaced regardless of label width, with their slant lines sized to fit the word. For modifiers with sub-modifiers, it is the *minimum* slant length. Increase it to widen spacing and push labels further from the baseline; decrease it to tighten compact diagrams.

“Deeper slants (`modDepth=1.6cm`)”

```

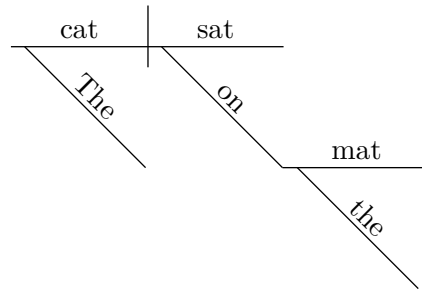
\begin{rkdiagram}[modDepth=1.6cm]

  \rkSubject[\rkAdj{The}]{cat}

  \rkVerb[\rkPrep[\rkAdj{the}]{on}{mat}]{sat}

\end{rkdiagram}

```



subClauseDepth — vertical drop to sub-baseline

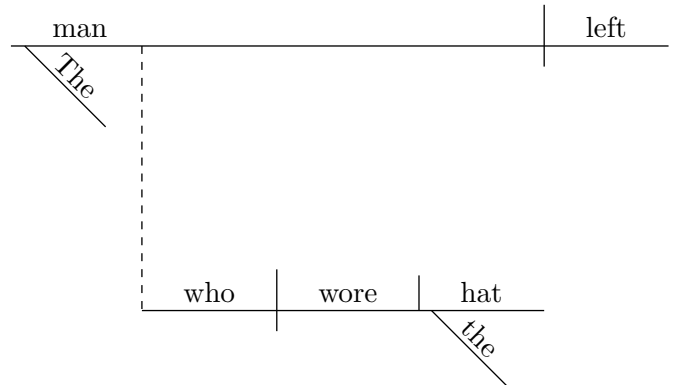
Controls how far below the main baseline a subordinate clause is drawn. Increase it when deep modifier chains on the main clause would overlap the sub-clause.

“Deeper sub-clause ($subClauseDepth=3.5cm$)”

```

\begin{rkdiagram}[subClauseDepth=3.5cm]
  \rkSubject[
    \rkAdj{The},
    \rkSubClause[who]{
      \rkVerb{wore}
      \rkDObj[\rkAdj{the}]{hat}
    }
  ]{man}
  \rkVerb{left}
\end{rkdiagram}

```



cmpSep — compound platform separation

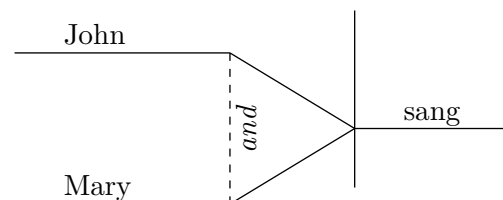
Sets the minimum vertical half-gap between the two platforms of a compound element. The default 0pt lets the package scale the gap automatically from the conjunction width. Set an explicit value when modifier chains on the compound words need extra vertical clearance.

“Extra separation ($cmpSep=1.0cm$)”

```

\begin{rkdiagram}[cmpSep=1.0cm]
  \rkSubject{
    \rkCompound[and]{John}{Mary}
  }
  \rkVerb{sang}
\end{rkdiagram}

```



bridgeY — connector bridge height (compound sentences)

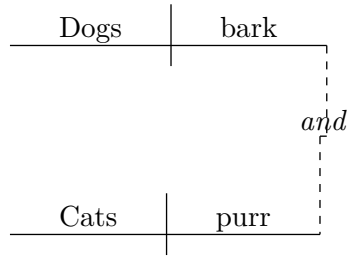
By default `\rkCompoundSent` places the horizontal bridge halfway between the two baselines, automatically lowering it if clause 1 modifiers extend into that region. Use `bridgeY` (a negative length, measured from clause 1's baseline) to pin the bridge at an exact position:

```
\rkCompoundSent[and][3cm][bridgeY=-1.8cm]
{\begin{rkdiagram}\rkSubject{Dogs}\rkVerb{bark}\end{rkdiagram}}
{\begin{rkdiagram}\rkSubject{Cats}\rkVerb{purr}\end{rkdiagram}}
```

Pass `bridgeY` in the first per-clause options argument. Setting it to `0pt` restores automatic placement.

“Bridge pinned at -1.2 cm ”

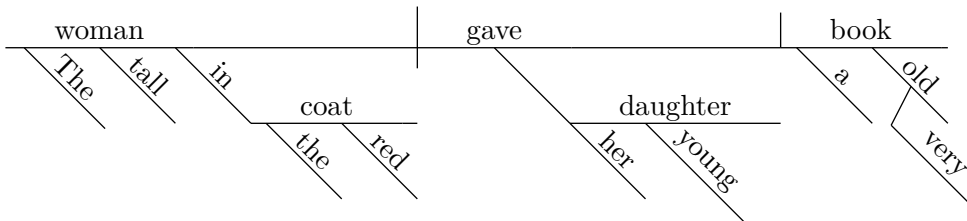
```
\rkCompoundSent[and][2.5cm]
[bridgeY=-1.2cm]
{\begin{rkdiagram}
\rkSubject{Dogs}\rkVerb{bark}
\end{rkdiagram}}
{\begin{rkdiagram}
\rkSubject{Cats}\rkVerb{purr}
\end{rkdiagram}}
```



Complete Example

“The tall woman in the red coat gave her young daughter a very old book.”

```
\begin{rkdiagram}
\rkSubject[
\rkAdj{The}, \rkAdj{tall},
\rkPrep[\rkAdj{the}, \rkAdj{red}]{in}{coat}
]{woman}
\rkVerb{gave}
\rkIObj[\rkAdj{her}, \rkAdj{young}]{daughter}
\rkDObj[\rkAdj{a}, \rkAdj[\rkAdv{very}]{old}]{book}
\end{rkdiagram}
```



Command Reference

Baseline commands

All baseline commands share the signature `\rkXxx[modifier-list]{word(s)}`. The modifier list is a comma-separated sequence of modifier, prep, appositive, or sub-clause calls.

Command	Role
<code>\rkSubject[mods]{word}</code>	Subject; left of the full vertical divider
<code>\rkVerb[mods]{word}</code>	Verb; right of the full vertical divider
<code>\rkDObj[mods]{word}</code>	Direct object; right of the short vertical bar
<code>\rkIObj[mods]{word}</code>	Indirect object; diagonal platform below verb
<code>\rkPredNom[mods]{word}</code>	Predicate nominative; right of the back-diagonal
<code>\rkPredAdj[mods]{word}</code>	Predicate adjective; right of the back-diagonal

Modifier and structural commands

Command	Role
<code>\rkAdj[mods]{word}</code>	Adjective/article; diagonal below modified noun
<code>\rkAdv[mods]{word}</code>	Adverb; diagonal below modified verb/adj/adv
<code>\rkPrep[obj-mods]{prep}{obj}</code>	Prep phrase; diagonal + horizontal object line
<code>\rkAppos{content}</code>	Appositive; parenthesized on the baseline
<code>\rkSubClause[rel]{body}</code>	Subordinate clause; dashed connector to sub-baseline
<code>\rkCompound[conj][mA]{wA}[mB]{wB}</code>	Compound element; forked platforms with conjunction
<code>\rkGerund[mods]{word}</code>	Gerund; stepped T-platform (used as word argument of any baseline command or <code>\rkPrep</code> object)
<code>\rkParticiple[tail]{word}</code>	Participle; hockey-stick modifier of a noun (used inside modifier lists like <code>\rkAdj</code>)
<code>\rkPartObj[mods]{word}</code>	Participial object on the tail; only inside <code>\rkParticiple</code> 's tail argument

Compound sentence

Command	Role
<code>\rkCompoundSent[conj][sep][opts1]{body1}[opts2]{body2}</code>	Two independent clauses (standalone; each wraps its own <code>rkdiagram</code>), connected by a bridge

Manual spacing (one-shot)

Command	Effect
<code>\rkSlotWidth{len}</code>	Minimum line width for the next baseline slot
<code>\rkSkip{len}</code>	Gap in baseline before the next slot
<code>\rkSlantLen{len}</code>	Minimum diagonal length for the next modifier/prep

Structural depth options (per-diagram or global)

These are standard package options (see §2) but can also be passed to any `rkdiagram` environment or to `\rkset` mid-document:

Option	Effect
<code>modDepth=<i>len</i></code>	Fixed step between plain adj/adv siblings; minimum slant length for sub-modified modifiers and preps
<code>subClauseDepth=<i>len</i></code>	Vertical drop from main baseline to sub-clause baseline
<code>cmpSep=<i>len</i></code>	Minimum half-gap between compound platforms (<code>0pt</code> = auto)
<code>bridgeY=<i>neg-len</i></code>	Pin compound-sentence bridge at a fixed y below clause 1 (<code>0pt</code> = auto)

Global configuration

Command	Effect
<code>\rkset{key=val,...}</code>	Override any package option mid-document

Compile with `lualatex`. Test suite: `rk-test.tex`.